Task 3 (cold-target): KIBA results and suggested changes

I ran your Task 3 plan on KIBA with the same settings (split rule, features, model, grid, metrics, seeds), so the two datasets can be compared. Below: what I had to change for KIBA, one extra model I added, the results, and a few things for us to agree on. Notebook: *notebooks/task3_kiba.ipynb*.

1. Kept exactly as in your plan

- Cold split by group: shuffled groups, cumulative row shares 0.70 / 0.85; seed 42 main run, seeds 0–4 repeats.
- 3-mer vocabulary and SVD (24 dims) fit on training sequences only, plus 20 amino-acid fractions and log length.
- Ligand mean leaving out the row's own group (training rows); Morgan fingerprint (r = 2, 1,024 count bits, RDKit) to SVD 16.
- HistGradientBoostingRegressor, min_samples_leaf = 40, the same 4-setting grid, chosen by validation MSE.
- ESM: facebook/esm2_t6_8M_UR50D, mean-pooled, 1,000-residue chunks, PCA 24 fit on train.
- Analyses: error vs. similarity (4 bins), permutation importance of feature groups, ESM vs. 3-mers, paired table.

2. What I changed for KIBA

Topic	DAVIS (your plan)	KIBA (what I did)
Data	68 × 442, complete	Task 1 cleaned table: 2,068 ligands × 229 targets, 117,657 observed pairs (stereoisomers merged). Only observed pairs used.
Target groups	gene name	No variants in KIBA, so I clustered *kiba_target_sim.txt* (average linkage, similarity > 0.5) → 150 groups, largest 8. Same families as Task 1 Fig. 7. No group or sequence crosses partitions on any split.
Score floor	clip at pKd 5; CI on pKd > 5	No clip. 23 % of scores pile up at 11.1 / 11.2, so I report **CI on pairs with both scores > 11.2** instead.
Direction	pKd	Higher KIBA score = stronger binding (DeepDTA version).
CI cost	full	CI on at most 20,000 test pairs, fixed seed.
Hit@k	per ligand	Only ligands with ≥ 5 held-out targets; ligands whose held-out scores are all equal are skipped. Chance level is high on KIBA (random Hit@5 ≈ 0.45).
Random baseline	random ranking	Random draws from the training scores (so its MSE is meaningful).

3. What I added: a similarity-neighbour model

Task 1 showed that similar kinases have similar binding profiles (median profile correlation 0.54 for the most similar pairs vs. 0.23). So for an unseen target *t*, I predict a ligand's score as the average of **that ligand's own training scores**, weighted by Smith–Waterman similarity to *t* raised to a power *q* (*q* in {1, 2, 4, 8, 16}, chosen on validation; it picks 2):

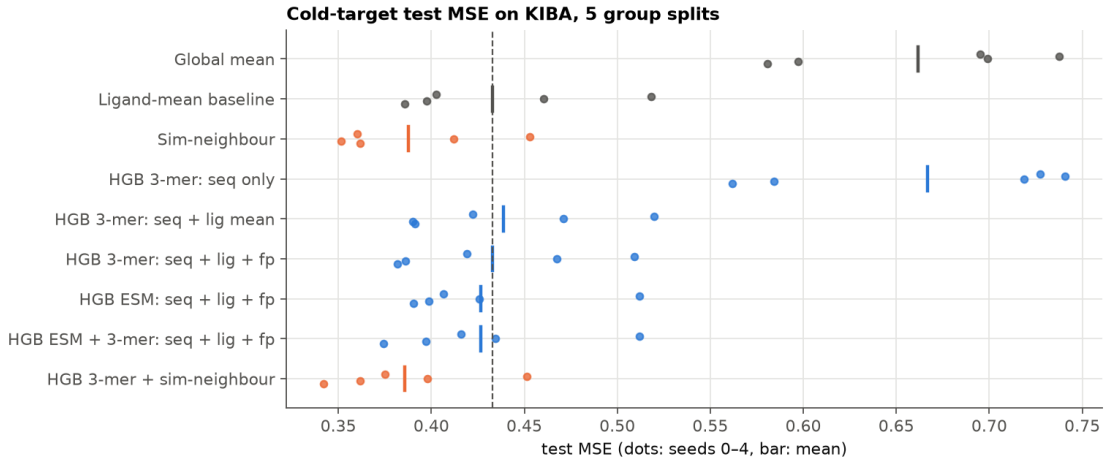
$$pred(l, t) = \Sigma\, s(t, t')^q \cdot y(l, t') \,/\, \Sigma\, s(t, t')^q \text{ over training targets } t' \text{ observed for ligand } l$$

Only sequences (via the supplied similarity matrix) and training scores are used, so it is still a fair cold-target method. For training rows, targets in the row's own group are excluded, mirroring your leave-group-out ligand mean. I use it (a) **on its own**, no training, and (b) as **two extra HGB features** (neighbour prediction + max similarity).

```
W = sim[:, train_targets] ** q            # zero where same group
pred = (S_train @ W.T) / (Mask_train @ W.T)  # ligands x targets; fall back to ligand mean
```

4. Results on KIBA (test set, 5 cold splits, seeds 0–4)

Model	MSE test	CI	CI > 11.2	Hit@5	Beats baseline (paired MSE)
Global mean	0.662	0.500	0.500	0.449	0 of 5
Ligand-mean baseline	0.433	0.699	0.656	0.449	(reference)
3-mer: seq only	0.667	0.552	0.565	0.534	0 of 5
3-mer: seq + ligand mean	0.439	0.696	0.662	0.506	1 of 5
3-mer: seq + lig + fp	0.433	0.698	0.662	0.498	3 of 5
ESM: seq + lig + fp	0.427	0.694	0.649	0.498	2 of 5
ESM + 3-mer: seq + lig + fp	0.427	0.700	0.656	0.515	4 of 5
[new] Sim-neighbour (no training)	**0.388**	**0.734**	**0.694**	**0.601**	**5 of 5**
[new] 3-mer + lig + fp + sim-neighbour	**0.386**	0.728	0.691	0.582	**5 of 5**



5. Takeaways

- **Your HGB models only match the ligand-mean baseline on KIBA** (mean ΔMSE –0.006 to 0.000; 2–4 of 5 seeds). Each split has only ~160 training kinases, too few to learn sequence → affinity. Train MSE 0.28 vs. validation 0.44 shows overfitting to seen targets.
- **The ligand mean carries most of the signal** (permutation importance 0.36 vs. 0.02 for 3-mers and 0.02 for the fingerprint), as on DAVIS.
- **The similarity-neighbour model wins on all 5 splits** (ΔMSE –0.045, CI +0.035, Hit@1 0.20 vs. 0.10). Each ligand has many measured kinases, so borrowing from related ones works. In HGB it becomes the dominant feature (importance 0.58).
- **The gain comes from test kinases with a related training kinase**: error is 18–24 % lower than the baseline in the upper three similarity bins, and only ~2 % lower in the least-similar bin.
- **ESM ≈ 3-mers** (ESM better on 2 of 5 KIBA splits; 3 of 5 on DAVIS). Mean-pooled ESM-2 8M does not add much on either dataset.

6. Suggestions and questions

- **Keep Task 3 separate from Task 2.** The CF model needs each target in training, so it cannot score unseen targets. We should share the Hit@k code with Task 2, though, so the metric is identical.
- **Keep your models and add the similarity-neighbour model**, and run it on DAVIS too so both datasets have the same model list. DAVIS has no similarity file; 3-mer cosine similarity (already computed for the error-vs-similarity plot) can replace it.
- **Question 1:** does “refit” in the plan mean refitting on train + validation? I scored the train-only model chosen on validation.
- **Question 2:** how does your Hit@k handle ties in the true scores (several targets with the best score)? I count a hit if any of them is in the top k.